

July 10

③. $g: \mathbb{R}^2 \rightarrow \mathbb{R}$ a C^1 fct.

↙ I see this form recurrently in a lot of exercises!

planar dif. system:
$$\begin{cases} \dot{x} = -y + x \cdot g(x,y) \\ \dot{y} = x + y \cdot g(x,y) \end{cases}$$

a) $(0,0)$ equil? other equil?

$$\begin{cases} -y + x \cdot g(x,y) = 0 & \Rightarrow y = x \cdot g(x,y) \\ x + y \cdot g(x,y) = 0 & \Rightarrow x + x \cdot g^2(x,y) = 0 \\ & \Rightarrow x(1 + \underbrace{g^2(x,y)}_{>0}) = 0 \quad (\Leftrightarrow) \quad x = 0 = y. \end{cases}$$

$\Rightarrow (0,0)$ unique equil.

b) lin. method; stability of $(0,0)$? discuss w.r.t. $g(0,0)$.

$f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $f(x,y) = \begin{pmatrix} -y + x \cdot g(x,y) \\ x + y \cdot g(x,y) \end{pmatrix}$

g is $C^1 \Rightarrow f$ is C^1 .

$Df(x,y) = \begin{pmatrix} x \cdot g(x,y) + x \cdot g'(x,y) & -1 + x \cdot g'(x,y) \\ 1 + y \cdot g'(x,y) & y \cdot g(x,y) + y \cdot g'(x,y) \end{pmatrix}$

correctly written $\begin{pmatrix} g(x,y) + x \cdot \frac{\partial g}{\partial x}(x,y) & -1 + x \cdot \frac{\partial g}{\partial y}(x,y) \\ 1 + y \cdot \frac{\partial g}{\partial x}(x,y) & g(x,y) + y \cdot \frac{\partial g}{\partial y}(x,y) \end{pmatrix}$

$Df(0,0) = \begin{pmatrix} g(0,0) & -1 \\ 1 & g(0,0) \end{pmatrix}$

$\begin{vmatrix} g(0,0) - \lambda & -1 \\ 1 & g(0,0) - \lambda \end{vmatrix} = 0 \Rightarrow (g(0,0) - \lambda)^2 + 1 = 0$
 $\Rightarrow g(0,0) - \lambda = \pm i$
 $\Rightarrow \lambda_{1,2} = g(0,0) \pm i$ PWS.

- If $g(0,0) \neq 0 \Rightarrow (0,0)$ hyperbolic.
- If $g(0,0) = 0 \Rightarrow$ lin. method ~~TAKES~~ $\Rightarrow$ we don't know stability.
- If $g(0,0) < 0 \Rightarrow (0,0)$ attractor., a. stable
- If $g(0,0) > 0 \Rightarrow$ repeller., a. unstable

STABLE $\Rightarrow$ GLOBAL ATTRACTOR

ASIMPTOTICALLY STABLE $\Rightarrow$ LOCAL ATTRACTOR (neighbourhood)

c) Prove that any orbit rotates around (0,0).

$\rightarrow \theta$ is monotone (strictly)

$$\begin{aligned} \tan \theta &= \frac{y}{x} \Rightarrow \frac{\dot{\theta}}{\cos^2 \theta} = \frac{\dot{y}x - y\dot{x}}{x^2} \quad \left. \begin{array}{l} x = \rho \cos \theta / 2 \\ \Downarrow \\ x^2 = \rho^2 \cos^2 \theta \\ \Rightarrow \cos^2 \theta = \frac{x^2}{\rho^2} \end{array} \right\} \Rightarrow \frac{\dot{\theta}}{\frac{x^2}{\rho^2}} = \frac{\dot{y}x - y\dot{x}}{x^2} \\ &\Rightarrow \frac{\rho^2 \cdot \dot{\theta}}{x^2} = \frac{\dot{y}x - y\dot{x}}{x^2} \\ &\Rightarrow \dot{\theta} = \frac{\dot{y}x - y\dot{x}}{\rho^2} \\ &\Rightarrow \dot{\theta} = \frac{1}{\rho^2} (\dots) = \frac{\rho^2}{\rho^2} = 1 > 0 \Rightarrow \text{angle } \theta(t) \text{ strictly increasing along any orbit.} \end{aligned}$$

d) In the case that g takes the value 0 in any point of the unit circle, check:

$$\varphi\left(t, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = \left(\cos\left(t + \frac{\pi}{4}\right), \sin\left(t + \frac{\pi}{4}\right)\right), \quad (\forall) t \in \mathbb{R}.$$

$$\begin{array}{ccccccc} & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ x(0) = \frac{1}{\sqrt{2}} & & y(0) = \frac{1}{\sqrt{2}} & & \tilde{x}(t) & & \tilde{y}(t) \end{array}$$

$$\begin{cases} \dot{x} = -y + x \cdot g(x, y) \\ \dot{y} = x + y \cdot g(x, y) \\ x(0) = \frac{1}{\sqrt{2}} \\ y(0) = \frac{1}{\sqrt{2}} \end{cases}$$

$$\tilde{x}(0) = \cos \frac{\pi}{4} = \frac{1}{\sqrt{2}}$$

$$\tilde{y}(0) = \sin \frac{\pi}{4} = \frac{1}{\sqrt{2}}$$

$$\tilde{x}(t)^2 + \tilde{y}(t)^2 = 1, \quad (\forall) t \in \mathbb{R} \Rightarrow (\tilde{x}(t), \tilde{y}(t)) \in \text{unit circle.} \Rightarrow$$

$$\Rightarrow g(\tilde{x}(t), \tilde{y}(t)) = 0$$

$$\tilde{x}'(t) = -\tilde{y}(t), \quad (\forall) t \in \mathbb{R}$$

$$\tilde{y}'(t) = \tilde{x}(t)$$

$$\Rightarrow (\tilde{x}(t), \tilde{y}(t)) \text{ satisfy.}$$

(iv) $a, b \in \mathbb{R}$, $f(x) = ax^2 + bx + 1$, $f(1) = 2$, $f(2) = 1$.

discrete scalar dym. sys. $x_{k+1} = f(x_k)$, $k \in \mathbb{N}$ $\rightarrow$ attr. 2-periodic orbit?

$f(1) = 2$, $f(2) = 1$

The orbit that starts at $\eta = 1$ is $\{1, 2\} \Rightarrow$ 2 periodic orbit.

$f(1) = 2 \Rightarrow a + b + 1 = 2 \Rightarrow a + b = 1 \Rightarrow b = 1 - a$

$f(2) = 1 \Rightarrow 4a + 2b + 1 = 1 \Rightarrow 2a + b = 0$

$2a + 1 - a = 0 \Rightarrow a = -1$

$b = 2$

$f(x) = -x^2 + 2x + 1$

$f'(x) = -2x + 2$

$|f'(\eta^*)| < 1 \Rightarrow$ attr.

$> 1 \Rightarrow$ ~~unstable~~ repeller, unstable

$f'(1) = 0$

$f'(2) = -2$

and $\Rightarrow |f'(1) \cdot f'(2)| = 0 < 1 \Rightarrow$ attr.